THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4

2. $\dfrac{3^6 \times 3^4}{3^5}$ equals: 1

(a) 3^5 **(b)** 3^7 **(c)** 3^{19} **(d)** 3^{15}

3. The decimal value of 10110_2 is: 1

(a) 20 **(b)** 24 **(c)** 28 **(d)** 22

4. The angles of a quadrilateral are x°, $(x+10)^\circ$, $(x+20)^\circ$ and $(x+30)^\circ$. The value of x is: 1

(a) 65 **(b)** 85 **(c)** 75 **(d)** 76

5. The digit that replaces * in 7*93 so that the number is divisible by 11 is: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 8

6. The smallest number by which 192 must be divided to get a perfect cube is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 6

7. The value of $\left(\dfrac{2}{3}\right)^{-2}$ is: 1

(a) 3 **(b)** $-\frac{9}{4}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{9}{4}$

8. The binary form of 45 is: 1

(a) 101110 **(b)** 110001 **(c)** 101101 **(d)** 101100

9. The adjacent sides of a parallelogram are in the ratio 3:5 and its perimeter is 64 cm. The longer side is: 1

(a) 32 cm **(b)** 40 cm **(c)** 20 cm **(d)** 12 cm

10. The smallest digit that replaces * in 52*8 so that the number is divisible by 6 is: 1

(a) 2 **(b)** 0 **(c)** 5 **(d)** 1

11. The ones digit of 127^2 is: 1

(a) 4 **(b)** 1 **(c)** 7 **(d)** 9

12. If $3^x \times 3^4 = 3^{11}$, then x is: 1

(a) 9 **(b)** 7 **(c)** 44 **(d)** 15

13. XLIV + LXVII equals: 1

(a) CI **(b)** CXII **(c)** CXXI **(d)** CXI

14. The diagonals of a parallelogram ABCD meet at O. If AO $= (2x + 3)$ cm and OC $= (3x - 4)$ cm, then AC is: 1

(a) 17 cm **(b)** 34 cm **(c)** 36 cm **(d)** 7 cm

15. One step of the Kaprekar routine applied to 1895 gives: 1

(a) 8361 **(b)** 8280 **(c)** 8163 **(d)** 8262

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find $\sqrt{2916}$ by prime factorisation. 2

17. Find the value of $(2^3)^3 \div 2^5$. 2

18. Convert 110101_2 to decimal. 2

19. The angles of a quadrilateral are in the ratio 2:3:5:8. Find the largest angle. 2

20. Find the smallest number to be added to 8456 to make it divisible by 11. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the smallest number by which 1872 must be multiplied to make it a perfect square. Find the square root of the product. 3

22. Simplify $\dfrac{2^5 \times 3^4}{2^3 \times 3^2}$ and find its value. 3

23. Convert 101101_2 to base 8 by first converting to decimal. 3

24. The angles of a quadrilateral are x°, $(x + 15)^\circ$, $(x + 30)^\circ$ and $(x + 45)^\circ$. Find all the angles. 3

OR

Find the smallest 4-digit number divisible by both 12 and 15. 3

25. Find the smallest number by which 12000 must be divided to make it a perfect cube. Find the cube root of the quotient. 3

OR

Evaluate $2^{-3} + 3^{-2}$ as a single fraction. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A school has 1500 students who are to stand in rows so that the number of rows equals the number of students in each row. How many more students are needed to complete the arrangement? What is the number of rows then? 5

OR

Simplify $\dfrac{2^3 \times 3^{-1}}{2^{-2} \times 3^1}$ and write the answer with positive exponents. Then find its value. 5

27. Find the sum of 1101_2, 1001_2 and 110_2. Verify the result in decimal. 5

OR

ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (2x + 10)^\circ$ and $\angle OBA = (3x - 5)^\circ$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

END OF QUESTION PAPER **Total: 50 Marks**

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